Supplementary Material

Does an Asymmetric-Reward E-DDQN Improve Safety-Prioritized Cybersickness Detection in Virtual Reality? A Leakage-Controlled Comparison with Standard DQN and Cost-Sensitive Baselines

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S1 Data, provenance and preprocessing

S1.1 Provenance

The evaluation data are the 35 sessions of the UFFCSData curated release (`DATABASE_SELECTED_DATA`, December 2019, seated posture; 9,390 samples at ≈1 Hz). Each curated file was matched sample by sample to one raw export (`RAW_DATABASE`); `dataset.csv`, the derived file used in the earlier version of this study, is exactly these sessions concatenated in lexicographic folder order (1, 10, 11, . . .), with integer-coded categories and no session identifier. The 37 October-2019 standing-posture raw sessions were used only for code development and budget calibration; their camera-orientation channels are constant within every session (a logging defect). Two December-2019 raw sessions that the dataset authors excluded were not used. Full per-session table: `results/preprocessing/session_table.csv`.

S1.2 Label

$y_t = 1$ iff the self-reported discomfort level (0–3) at second t is ≥ 1. Prevalence 44.2 % (Flight 41.2 %, Race 48.1 %); 9 of 35 sessions never report discomfort; the median onset is 82 s after the start. Labels are close to step functions within a session (on average 1.1 binary switches per session), so consecutive samples are strongly dependent.

S1.3 Linkage groups

No participant identifier exists. Sessions are linked when their nine-field questionnaire profile is identical or when one restarts within 180 s after an aborted session (< 200 samples) with at most two differing fields; this gives 21 groups (log: `results/preprocessing/linkage_log.csv`). Groups are the unit of every split.

S1.4 Causal features

Definitions of all 39 telemetry features and 11 questionnaire indicators: `results/preprocessing/feature_dictionary.csv`. The models use preprocessing (imputation value, spike threshold, scaler centre and scale) fitted on the training rows of each inner split. For documentation, `learned_parameters_by_outer_fold.csv` lists the same parameters fitted on each outer fold's development groups; no test rows enter either fit. Executable checks: truncation invariance of all features (max. difference 0.0 over 40 random cut points); bit-identical preprocessing parameters when test rows are tampered with; zero session, group, identical-profile and duplicated-feature-row overlap between every development set and its test fold (`partition_overlap_checks.csv`).

S1.5 Removed variables

Field of view, static-frame and auto-camera flags (constant within each game: duplicates of the game indicator); haptic feedback, degree of control, DOF simulation, locomotion, posture and symptoms (constant overall); absolute player position (encodes course distance and therefore elapsed time; absent in the October wave); VRSQ scores (post-session).

S2 Protocol, search spaces and selected configurations

The protocol (`configs/protocol.json`) and the perturbation configuration (`configs/attacks.json`) were committed to the repository before any model was trained on the evaluation sessions (commit recorded in `configs/FROZEN_PROTOCOL_COMMIT.txt`). Outer-test rows are not even featurised during hyper-parameter search; the final stage scores each outer-test fold once.

S2.1 Search spaces

Family	Hyper-parameter	Candidate values
All neural models	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
All neural models	hidden	[64, 64], [128, 128], [256, 256], [256, 512, 256]
All neural models	dropout	0.0, 0.1, 0.2, 0.3
All neural models	norm	batch, layer
All neural models	lr	0.0003, 0.001, 0.003
All neural models	weight_decay	0.0, 0.0001, 0.001
All neural models	batch_size	256, 512

Family	Hyper-parameter	Candidate values
All agents	eps_decay	0.9, 0.95, 0.97
All agents	warmup	0, 4000
All agents	updates_per_learn	1, 2, 4
E-DDQN	gamma	0.0, 0.5, 0.9, 0.99
E-DDQN	tau	0.005, 0.01, 0.05
E-DDQN	rho	0.4, 0.5, 0.6
E-DDQN	full_info	False, True
E-CB	rho	0.4, 0.5, 0.6
E-CB	full_info	False, True
DQN	gamma	0.0, 0.5, 0.9, 0.99
DQN	hard_every	10, 100, 500
CS-MLP	sampling	uniform, balanced
LogReg	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
LogReg	C	0.0001, 0.001, 0.01, 0.1, 1.0, 10.0
HGB	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
HGB	learning_rate	0.02, 0.05, 0.1
HGB	max_iter	50, 100, 200, 400
HGB	max_leaf_nodes	4, 8, 16, 31
HGB	min_samples_leaf	20, 100, 300
HGB	l2_regularization	0.0, 1.0, 10.0
RF	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
RF	max_depth	3, 5, 8, None
RF	min_samples_leaf	5, 25, 100, 300
RF	max_features	sqrt, 0.5, 1.0
RF	n_estimators	300
All agents	fixed	epochs=40, eval_every=2, chunk=256, learn_every=2, consolidation=5, eps_start=1.0, eps_end=0.01, buffer=50000, grad_clip=1.0

Number of distinct configurations per family: E-DDQN 2,985,984, DQN 497,664, E-CB 248,832, CS-MLP 4,608, HGB 1,728, random forest 192, logistic regression 24. Each family received 16 random configurations (12 for logistic regression), so the budget is matched in number, not in coverage.

S2.2 Selected configuration per outer fold

Selection criterion: pooled inner out-of-fold 5:1 expected cost at the cost-minimising threshold (ties: PR-AUC).

Model	Outer	OOF cost	Selected configuration
E-DDQN	0	0.424	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	1	0.452	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	2	0.484	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	3	0.490	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=False, rho=0.4, tau=0.005, norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.0001, batch_size=512, eps_decay=0.9, warmup=0, updates_per_learn=1
E-DDQN	4	0.483	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.9, full_info=False, rho=0.5, tau=0.05, norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.001, batch_size=512, eps_decay=0.95, warmup=4000, updates_per_learn=2
E-CB	0	0.472	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	1	0.459	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	2	0.518	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4

Model	Outer	OOF cost	Selected configuration
E-CB	3	0.523	feature_set=exposure, hidden=[128, 128], gamma=0.0, full_info=False, rho=0.6, norm=layer, dropout=0.2, lr=0.001, weight_decay=0.001, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=2
E-CB	4	0.466	feature_set=exposure+profile, hidden=[64, 64], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.001, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=1
DQN	0	0.432	feature_set=exposure, hidden=[64, 64], gamma=0.0, full_info=False, hard_every=10, norm=layer, dropout=0.0, lr=0.001, weight_decay=0.0001, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=2
DQN	1	0.452	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.9, full_info=False, hard_every=500, norm=layer, dropout=0.0, lr=0.001, weight_decay=0.001, batch_size=512, eps_decay=0.9, warmup=0, updates_per_learn=4
DQN	2	0.538	feature_set=exposure, hidden=[128, 128], gamma=0.9, full_info=False, hard_every=500, norm=batch, dropout=0.3, lr=0.0003, weight_decay=0.0, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=2
DQN	3	0.501	feature_set=exposure, hidden=[128, 128], gamma=0.9, full_info=False, hard_every=100, norm=layer, dropout=0.2, lr=0.001, weight_decay=0.0, batch_size=512, eps_decay=0.97, warmup=0, updates_per_learn=2
DQN	4	0.469	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.0, full_info=False, hard_every=100, norm=batch, dropout=0.1, lr=0.003, weight_decay=0.0, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=1
CS-MLP	0	0.485	feature_set=exposure+profile, hidden=[128, 128], norm=layer, dropout=0.3, lr=0.001, weight_decay=0.0, batch_size=256, sampling=balanced
CS-MLP	1	0.490	feature_set=exposure+profile, hidden=[128, 128], norm=layer, dropout=0.3, lr=0.001, weight_decay=0.0, batch_size=256, sampling=balanced
CS-MLP	2	0.503	feature_set=exposure, hidden=[64, 64], norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.0, batch_size=256, sampling=uniform
CS-MLP	3	0.527	feature_set=telemetry, hidden=[64, 64], norm=layer, dropout=0.0, lr=0.003, weight_decay=0.0001, batch_size=256, sampling=balanced
CS-MLP	4	0.500	feature_set=telemetry, hidden=[64, 64], norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.001, batch_size=512, sampling=uniform
LogReg	0	0.486	feature_set=exposure, C=0.0001
LogReg	1	0.522	feature_set=telemetry+profile, C=1.0
LogReg	2	0.540	feature_set=exposure, C=10.0
LogReg	3	0.511	feature_set=exposure, C=0.0001
LogReg	4	0.531	feature_set=exposure+profile, C=0.01
HGB	0	0.486	feature_set=exposure, learning_rate=0.02, max_iter=50, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=1.0
HGB	1	0.501	feature_set=exposure, learning_rate=0.02, max_iter=50, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=1.0
HGB	2	0.542	feature_set=exposure, learning_rate=0.02, max_iter=400, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=0.0
HGB	3	0.532	feature_set=telemetry, learning_rate=0.05, max_iter=200, max_leaf_nodes=31, min_samples_leaf=100, l2_regularization=0.0
HGB	4	0.527	feature_set=exposure, learning_rate=0.02, max_iter=400, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=0.0
RF	0	0.465	feature_set=telemetry, min_samples_leaf=100, max_depth=3, max_features=0.5
RF	1	0.478	feature_set=exposure, min_samples_leaf=25, max_depth=8, max_features=sqrt
RF	2	0.509	feature_set=exposure, min_samples_leaf=5, max_depth=5, max_features=sqrt
RF	3	0.514	feature_set=exposure, min_samples_leaf=5, max_depth=5, max_features=sqrt
RF	4	0.510	feature_set=telemetry+profile, min_samples_leaf=25, max_depth=3, max_features=sqrt

S3 Additional results

S3.1 Full metric panel (primary rule D1)

Mean ± SD over five seeds (%, except MCC and cost); TP/TN/FP/FN are the exact confusion-matrix counts of the seed-ensemble predictions on all 9,390 outer-test samples. FPR = 1−specificity and FNR = 1−recall.

Model	Acc.	Prec.	Rec.	Spec.	F1	Bal.acc.	MCC	G-mean	FPR	FNR	$\text{Cost}_{5:1}$	TP	TN	FP	FN
E-DDQN	45.4±1.3	44.4±0.7	93.4±2.5	7.4±2.7	60.2±0.9	50.4±1.2	0.016±0.054	26.0±4.5	92.6±2.7	6.6±2.5	0.663±0.048	3,747	423	4,820	400
E-CB	47.6±1.2	45.6±0.6	97.1±2.9	8.4±2.8	62.1±1.0	52.8±1.2	0.122±0.052	28.2±5.0	91.6±2.8	2.9±2.9	0.575±0.055	4,007	763	4,480	140
DQN	50.1±1.9	46.8±1.0	95.2±2.7	14.5±4.9	62.8±0.8	54.8±1.6	0.158±0.038	36.6±5.8	85.5±4.9	4.8±2.7	0.584±0.040	4,018	852	4,391	129
CS-MLP	45.4±0.9	44.7±0.4	99.4±0.7	2.8±1.9	61.7±0.4	51.1±0.8	0.072±0.049	14.7±8.8	97.2±1.9	0.6±0.7	0.557±0.012	4,124	318	4,925	23
LogReg	48.3±0.0	46.0±0.0	99.1±0.0	8.1±0.0	62.9±0.0	53.6±0.0	0.167±0.000	28.4±0.0	91.9±0.0	0.9±0.0	0.532±0.000	4,111	426	4,817	36
HGB	51.0±0.0	47.3±0.0	97.0±0.0	14.6±0.0	63.6±0.0	55.8±0.0	0.197±0.000	37.6±0.0	85.4±0.0	3.0±0.0	0.543±0.000	4,023	764	4,479	124
RF	53.6±0.3	48.8±0.2	98.5±0.3	18.2±0.7	65.2±0.1	58.3±0.3	0.266±0.004	42.3±0.8	81.8±0.7	1.5±0.3	0.491±0.003	4,093	962	4,281	54

S3.2 Secondary operating rules

Rule D0 (untuned Bayes cut-off $\hat{p} \geq 1/6$; for the asymmetric agents this is the greedy policy):

Model	Recall	Spec.	Prec.	F1	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	FN	FP
E-DDQN	77.2±3.8	28.0±2.0	45.9±1.7	57.6±2.4	52.6±2.7	0.060±0.061	0.905±0.092	0.538±0.016	0.483±0.019	909	3,862
E-CB	85.9±2.2	37.2±3.0	52.0±1.1	64.7±1.1	61.5±1.5	0.258±0.031	0.663±0.044	0.653±0.011	0.562±0.006	627	3,262
DQN	85.1±1.7	37.1±1.5	51.7±0.7	64.3±0.9	61.1±1.0	0.248±0.022	0.680±0.035	0.661±0.013	0.595±0.022	561	3,330
CS-MLP	83.0±3.1	36.4±2.3	50.8±1.0	63.0±1.5	59.7±1.6	0.215±0.037	0.731±0.066	0.644±0.018	0.521±0.022	622	3,381
LogReg	85.3±0.0	13.6±0.0	43.8±0.0	57.9±0.0	49.4±0.0	−0.016±0.000	0.808±0.000	0.626±0.000	0.576±0.000	611	4,531
HGB	92.0±0.0	30.0±0.0	51.0±0.0	65.6±0.0	61.0±0.0	0.272±0.000	0.567±0.000	0.692±0.000	0.593±0.000	331	3,670
RF	94.5±0.1	34.9±0.1	53.4±0.0	68.3±0.0	64.7±0.0	0.353±0.001	**0.486±0.001**	0.688±0.002	0.548±0.001	228	3,410
Always alarm	100.0	0.0	44.2	61.3	50.0	0.000	0.558	–	–	0	5,243
Never alarm	0.0	100.0	–	–	50.0	0.000	2.208	–	–	4,147	0

Validation recall constraint ($\geq$0.95 on pooled inner-OOF predictions):

Model	Recall	Spec.	Prec.	F1	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	FN	FP
E-DDQN	86.8±3.1	11.5±3.4	43.7±0.9	58.1±1.3	49.1±1.7	−0.027±0.053	0.785±0.062	0.538±0.016	0.483±0.019	641	4,539
E-CB	95.2±2.7	15.0±3.3	47.0±0.8	62.9±1.0	55.1±1.4	0.167±0.042	0.581±0.051	0.653±0.011	0.562±0.006	317	4,093
DQN	93.5±2.9	23.0±3.4	49.0±0.8	64.3±1.0	58.3±1.2	0.227±0.032	0.573±0.052	0.661±0.013	0.595±0.022	343	3,810
CS-MLP	98.3±1.2	9.3±4.2	46.2±1.0	62.8±0.9	53.8±1.9	0.156±0.052	0.544±0.025	0.644±0.018	0.521±0.022	105	4,463
LogReg	97.2±0.0	18.3±0.0	48.5±0.0	64.7±0.0	57.7±0.0	0.241±0.000	0.518±0.000	0.626±0.000	0.576±0.000	116	4,285
HGB	97.5±0.0	18.3±0.0	48.6±0.0	64.8±0.0	57.9±0.0	0.248±0.000	0.511±0.000	0.692±0.000	0.593±0.000	103	4,282
RF	97.0±0.2	25.4±0.3	50.7±0.0	66.6±0.0	61.2±0.0	0.307±0.002	**0.483±0.004**	0.688±0.002	0.548±0.001	128	3,896
Always alarm	100.0	0.0	44.2	61.3	50.0	0.000	0.558	–	–	0	5,243
Never alarm	0.0	100.0	–	–	50.0	0.000	2.208	–	–	4,147	0

S3.3 Group-bootstrap confidence intervals

Seed-ensemble predictions, rule D1; 95 % percentile intervals from 2,000 bootstrap replicates of the 21 linkage groups.

Model (seed ensemble, D1)	Recall	Spec.	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC
E-DDQN	90.4 [77.7, 98.9]	8.1 [1.5, 17.6]	0.726 [0.533, 0.996]	0.541 [0.382, 0.690]	0.496 [0.329, 0.683]
E-CB	96.6 [92.2, 99.6]	14.6 [8.3, 21.4]	0.552 [0.431, 0.673]	0.659 [0.550, 0.767]	0.565 [0.429, 0.740]
DQN	96.9 [92.6, 99.7]	16.3 [8.7, 24.8]	0.536 [0.417, 0.654]	0.665 [0.558, 0.785]	0.606 [0.456, 0.778]
CS-MLP	99.4 [98.5, 100.0]	6.1 [2.6, 10.8]	0.537 [0.421, 0.634]	0.654 [0.527, 0.781]	0.529 [0.393, 0.712]
LogReg	99.1 [98.0, 100.0]	8.1 [4.0, 13.0]	0.532 [0.429, 0.627]	0.626 [0.498, 0.750]	0.576 [0.408, 0.754]
HGB	97.0 [92.4, 99.9]	14.6 [7.0, 24.9]	0.543 [0.439, 0.654]	0.692 [0.574, 0.809]	0.593 [0.437, 0.779]
RF	98.7 [97.5, 99.6]	18.3 [11.7, 28.0]	0.485 [0.381, 0.577]	0.688 [0.582, 0.808]	0.548 [0.443, 0.735]

S3.4 Per-fold results

Expected 5:1 cost / ROC-AUC of the seed ensemble in each outer fold (rule D1). Fold-to-fold variation is large because each fold contains only 3–5 linkage groups.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
E-DDQN	0.604 / 0.536	0.762 / 0.522	0.529 / 0.734	1.184 / 0.501	0.693 / 0.466
E-CB	0.569 / 0.655	0.537 / 0.718	0.462 / 0.756	0.460 / 0.747	0.758 / 0.447
DQN	0.564 / 0.715	0.757 / 0.521	0.451 / 0.852	0.426 / 0.740	0.520 / 0.499
CS-MLP	0.604 / 0.575	0.542 / 0.596	0.495 / 0.820	0.528 / 0.708	0.501 / 0.659
LogReg	0.582 / 0.623	0.542 / 0.428	0.486 / 0.810	0.531 / 0.712	0.515 / 0.610
HGB	0.619 / 0.631	0.547 / 0.740	0.510 / 0.791	0.522 / 0.761	0.495 / 0.585
RF	0.600 / 0.630	0.463 / 0.728	0.397 / 0.771	0.457 / 0.761	0.486 / 0.685

S3.5 Event-level behavior

Model	Events detected	Median delay (s)	Alarm % before onset	Alarm % in event-free sessions
E-DDQN	26/26	0	82.3	96.6
E-CB	25/26	0	73.1	76.9
DQN	25/26	0	77.4	70.4
CS-MLP	26/26	0	92.9	88.1
LogReg	25/26	0	76.0	91.7
HGB	25/26	0	69.3	90.8
RF	25/26	0	61.3	75.3

S3.6 Paired tests under the secondary rules

Rule D0:

B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\mathrm{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	635	1,517	4,771	3,889	360.7	<0.001	+0.214 [+0.025, +0.487]	0.018
DQN	467	1,347	4,771	3,891	425.9	<0.001	+0.242 [+0.039, +0.542]	0.012
CS-MLP	431	1,199	4,771	4,003	360.9	<0.001	+0.204 [−0.030, +0.507]	0.11
LogReg	1,297	926	4,771	5,142	61.6	<0.001	+0.087 [−0.293, +0.541]	0.69
HGB	833	1,603	4,771	4,001	242.8	<0.001	+0.328 [+0.078, +0.639]	0.003
RF	526	1,659	4,771	3,638	586.5	<0.001	+0.411 [+0.147, +0.748]	0.001

Validation recall ≥ 0.95:

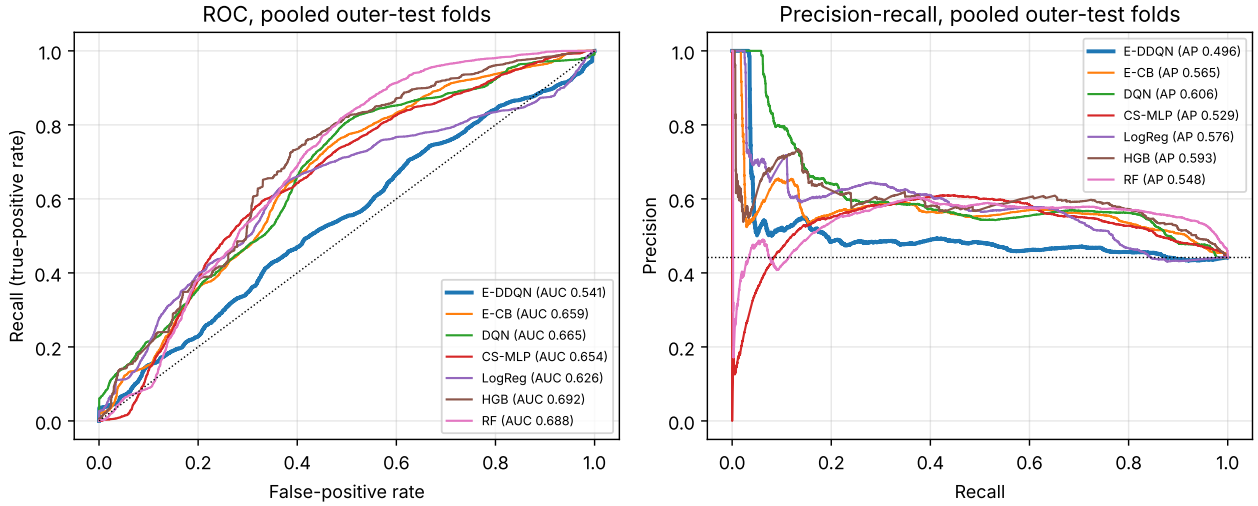
B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\mathrm{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	378	1,148	5,180	4,410	387.5	<0.001	+0.220 [+0.017, +0.566]	0.021
DQN	294	1,321	5,180	4,153	651.8	<0.001	+0.236 [+0.013, +0.580]	0.028
CS-MLP	509	1,121	5,180	4,568	229.0	<0.001	+0.294 [+0.033, +0.652]	0.013
LogReg	538	1,317	5,180	4,401	326.3	<0.001	+0.307 [+0.027, +0.687]	0.02
HGB	434	1,229	5,180	4,385	379.1	<0.001	+0.314 [+0.030, +0.711]	0.023
RF	393	1,549	5,180	4,024	686.9	<0.001	+0.342 [+0.067, +0.719]	0.003

S3.7 Value-estimation diagnostics

Mean Q-value of the greedy action on the inner-validation groups compared with the discounted return actually obtained by the greedy policy on those sessions (rewards of every action are known offline).

Model	γ (selected)	mean $Q(s, \pi(s))$	mean MC return	bias (mean ± SD over runs)
DQN	0–0.9	1.467	1.074	+0.393 ± 1.493
E-CB	0–0	0.224	0.136	+0.088 ± 0.078
E-DDQN	0.9–0.99	5.998	8.898	−2.900 ± 8.250

S3.8 ROC and precision–recall curves (primary models)



S4 Telemetry-perturbation stress tests

S4.1 Design (frozen before evaluation; `configs/attacks.json`)

Perturbations act on the raw per-second channels of the outer-test sessions; causal features are then recomputed with the training-fitted featurizer and scaler of each inner model. Cybersickness labels are never modified; the indicator z marks perturbed seconds. Stress scenarios (each over the whole session or one random 60-s window): frame-rate cap at 72, 60 or 45 fps (Type I); uniform orientation jitter of up to ± 0.02, 0.05 or 0.10 per rotation component (Type II); speed up-scaling by 10, 25 or 50 % with acceleration recomputed as the per-second speed difference (the logged acceleration equals this difference in every session; Type II); speed raised by 10, 20 or 40 units for one second every 20 s (Type II). Together: 4 kinds $\times$ 3 strengths $\times$ 2 windows $=$ 24 stress variants. Evasion-only candidates (whole session): speed down-scaling by 25 or 50 %, orientation smoothing with $\lambda = 0.5$ or 0.25, frame-rate restoration to 90 fps (5 variants; 29 variants in total, 17 of them whole-session). All values are clipped to the per-game ranges observed in the data. Field-of-view manipulation is not evaluated because the field of view never varies within a game in the data.

S4.2 Interpretation limits

The tests measure how bounded manipulation of the logged channels moves the detectors' scores and decisions. They do not show that a manipulation would cause or hide cybersickness in a person, because no physiological effect was measured. The adaptive worst cases give the adversary the labels and choose the most damaging candidate per session and model.

S4.3 Operating points and adaptive worst cases

Recall/false-positive rate (%) of the seed ensembles on perturbed seconds at the D1 operating point for representative whole-session variants, and per-session adaptive worst cases (evasion: minimum recall; false-alarm induction: maximum FPR) over all 17 whole-session candidates.

Model	Clean Rec./FPR	fps≤45 Rec./FPR	jitter 0.10 Rec./FPR	speed ×1.5 Rec./FPR	spikes 40 Rec./FPR	Adaptive worst case Rec. (evasion)	Adaptive worst case FPR (false alarm)
E-DDQN	90.4/91.9	90.4/91.9	94.0/92.2	93.5/90.7	95.4/94.3	82.9	94.5
E-CB	96.6/85.4	96.6/85.4	96.1/88.2	98.7/95.9	99.3/97.7	78.0	98.9
DQN	96.9/83.7	96.9/83.7	94.8/83.5	97.8/91.8	97.9/95.4	87.3	97.5
CS-MLP	99.4/93.9	100.0/98.3	99.7/96.5	100.0/98.2	99.9/99.1	95.6	100.0
LogReg	99.1/91.9	97.0/86.3	99.2/92.7	99.7/94.9	99.9/96.7	95.4	96.7
HGB	97.0/85.4	97.0/85.9	96.3/82.7	98.2/91.7	98.1/95.7	81.0	96.1
RF	98.7/81.7	98.7/81.3	98.0/81.1	99.0/92.1	99.8/97.4	90.5	98.4

S4.4 All variants

Recall and false-positive rate on perturbed seconds (%) and 5:1 cost on those seconds, clean vs. perturbed, seed ensembles at the D1 operating point.

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
CS-MLP	accel spikes	10	whole	99.4	99.6	93.9	95.5	0.537	0.542
CS-MLP	accel spikes	10	window60	99.3	99.7	90.5	92.4	0.532	0.536
CS-MLP	accel spikes	20	whole	99.4	99.8	93.9	98.3	0.537	0.553
CS-MLP	accel spikes	20	window60	100.0	100.0	89.2	97.9	0.545	0.599
CS-MLP	accel spikes	40	whole	99.4	99.9	93.9	99.1	0.537	0.556
CS-MLP	accel spikes	40	window60	100.0	100.0	97.0	99.6	0.496	0.510
CS-MLP	fps cap	45	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	45	window60	99.5	100.0	95.4	98.8	0.509	0.514
CS-MLP	fps cap	60	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	60	window60	98.6	100.0	95.4	99.0	0.649	0.649
CS-MLP	fps cap	72	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	72	window60	99.5	100.0	96.6	98.8	0.511	0.510
CS-MLP	fps restore	90	whole	99.4	99.7	93.9	93.7	0.537	0.530
CS-MLP	rot jitter	0.02	whole	99.4	99.1	93.9	92.0	0.537	0.534
CS-MLP	rot jitter	0.02	window60	99.9	99.7	96.7	94.6	0.517	0.511
CS-MLP	rot jitter	0.05	whole	99.4	98.8	93.9	91.4	0.537	0.536
CS-MLP	rot jitter	0.05	window60	98.3	96.4	94.4	91.2	0.656	0.667
CS-MLP	rot jitter	0.1	whole	99.4	99.7	93.9	96.5	0.537	0.545
CS-MLP	rot jitter	0.1	window60	100.0	99.8	94.3	96.2	0.560	0.576
CS-MLP	rot smooth	0.25	whole	99.4	99.5	93.9	97.1	0.537	0.553
CS-MLP	rot smooth	0.5	whole	99.4	99.3	93.9	96.8	0.537	0.555
CS-MLP	speed scale down	−0.5	whole	99.4	95.6	93.9	85.3	0.537	0.573
CS-MLP	speed scale down	−0.25	whole	99.4	98.5	93.9	90.2	0.537	0.538
CS-MLP	speed scale up	0.1	whole	99.4	99.6	93.9	95.1	0.537	0.540
CS-MLP	speed scale up	0.1	window60	99.3	99.7	92.2	94.1	0.541	0.544
CS-MLP	speed scale up	0.25	whole	99.4	100.0	93.9	96.7	0.537	0.541
CS-MLP	speed scale up	0.25	window60	100.0	100.0	95.3	97.2	0.534	0.544
CS-MLP	speed scale up	0.5	whole	99.4	100.0	93.9	98.2	0.537	0.548
CS-MLP	speed scale up	0.5	window60	100.0	100.0	87.4	97.1	0.448	0.497
DQN	accel spikes	10	whole	96.9	96.8	83.7	86.1	0.536	0.552
DQN	accel spikes	10	window60	92.6	92.6	80.3	82.7	0.617	0.631
DQN	accel spikes	20	whole	96.9	98.3	83.7	95.5	0.536	0.570
DQN	accel spikes	20	window60	96.9	97.4	78.5	98.0	0.540	0.650
DQN	accel spikes	40	whole	96.9	97.9	83.7	95.4	0.536	0.578
DQN	accel spikes	40	window60	99.5	100.0	87.5	99.8	0.460	0.511
DQN	fps cap	45	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	45	window60	94.1	94.1	84.5	84.5	0.581	0.581
DQN	fps cap	60	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	60	window60	94.5	94.5	83.7	83.7	0.643	0.643
DQN	fps cap	72	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	72	window60	93.5	93.5	83.9	83.9	0.591	0.591
DQN	fps restore	90	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	rot jitter	0.02	whole	96.9	94.2	83.7	78.0	0.536	0.563
DQN	rot jitter	0.02	window60	95.6	95.5	85.1	83.4	0.555	0.548
DQN	rot jitter	0.05	whole	96.9	93.2	83.7	74.7	0.536	0.568
DQN	rot jitter	0.05	window60	92.5	78.9	82.7	74.8	0.676	0.852
DQN	rot jitter	0.1	whole	96.9	94.8	83.7	83.5	0.536	0.581
DQN	rot jitter	0.1	window60	96.3	95.7	86.6	84.7	0.590	0.591
DQN	rot smooth	0.25	whole	96.9	99.6	83.7	92.4	0.536	0.525
DQN	rot smooth	0.5	whole	96.9	98.9	83.7	91.0	0.536	0.533
DQN	speed scale down	−0.5	whole	96.9	92.2	83.7	70.6	0.536	0.567
DQN	speed scale down	−0.25	whole	96.9	94.7	83.7	76.9	0.536	0.547
DQN	speed scale up	0.1	whole	96.9	97.0	83.7	88.1	0.536	0.557
DQN	speed scale up	0.1	window60	92.6	92.6	84.6	86.5	0.640	0.651
DQN	speed scale up	0.25	whole	96.9	97.4	83.7	90.8	0.536	0.564
DQN	speed scale up	0.25	window60	97.2	97.0	82.6	87.9	0.523	0.558
DQN	speed scale up	0.5	whole	96.9	97.8	83.7	91.8	0.536	0.562
DQN	speed scale up	0.5	window60	99.5	99.5	76.4	86.4	0.403	0.455
E-CB	accel spikes	10	whole	96.6	97.9	85.4	88.3	0.552	0.540
E-CB	accel spikes	10	window60	97.0	98.2	77.4	79.1	0.506	0.491
E-CB	accel spikes	20	whole	96.6	98.9	85.4	97.0	0.552	0.566
E-CB	accel spikes	20	window60	95.0	96.0	76.4	94.8	0.564	0.657
E-CB	accel spikes	40	whole	96.6	99.3	85.4	97.7	0.552	0.562
E-CB	accel spikes	40	window60	99.5	100.0	89.4	99.3	0.469	0.508
E-CB	fps cap	45	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	45	window60	96.0	96.0	84.2	84.2	0.533	0.533
E-CB	fps cap	60	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	60	window60	94.6	94.6	87.2	87.2	0.663	0.663
E-CB	fps cap	72	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	72	window60	97.0	97.0	86.2	86.2	0.518	0.518
E-CB	fps restore	90	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	rot jitter	0.02	whole	96.6	96.5	85.4	76.3	0.552	0.504
E-CB	rot jitter	0.02	window60	97.9	95.5	85.3	81.9	0.502	0.540
E-CB	rot jitter	0.05	whole	96.6	94.4	85.4	79.1	0.552	0.565
E-CB	rot jitter	0.05	window60	93.9	86.8	84.6	71.4	0.664	0.695
E-CB	rot jitter	0.1	whole	96.6	96.1	85.4	88.2	0.552	0.578
E-CB	rot jitter	0.1	window60	97.7	95.0	88.5	82.7	0.572	0.593
E-CB	rot smooth	0.25	whole	96.6	95.9	85.4	94.4	0.552	0.618
E-CB	rot smooth	0.5	whole	96.6	95.3	85.4	90.9	0.552	0.612
E-CB	speed scale down	−0.5	whole	96.6	81.0	85.4	59.5	0.552	0.752
E-CB	speed scale down	−0.25	whole	96.6	88.9	85.4	71.2	0.552	0.643
E-CB	speed scale up	0.1	whole	96.6	98.0	85.4	91.0	0.552	0.552
E-CB	speed scale up	0.1	window60	97.4	98.2	79.4	82.2	0.509	0.508
E-CB	speed scale up	0.25	whole	96.6	98.2	85.4	94.1	0.552	0.565
E-CB	speed scale up	0.25	window60	95.6	95.4	84.8	91.4	0.572	0.614
E-CB	speed scale up	0.5	whole	96.6	98.7	85.4	95.9	0.552	0.564
E-CB	speed scale up	0.5	window60	99.1	99.5	73.3	89.3	0.397	0.469
E-DDQN	accel spikes	10	whole	90.4	90.5	91.9	92.1	0.726	0.723
E-DDQN	accel spikes	10	window60	91.7	91.4	97.9	98.0	0.737	0.745

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
E-DDQN	accel spikes	20	whole	90.4	94.9	91.9	93.8	0.726	0.637
E-DDQN	accel spikes	20	window60	88.8	90.4	94.5	96.0	0.796	0.774
E-DDQN	accel spikes	40	whole	90.4	95.4	91.9	94.3	0.726	0.627
E-DDQN	accel spikes	40	window60	94.0	94.0	98.7	99.9	0.651	0.657
E-DDQN	fps cap	45	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	45	window60	90.2	90.2	99.3	99.3	0.752	0.752
E-DDQN	fps cap	60	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	60	window60	83.7	83.7	95.5	95.5	0.907	0.907
E-DDQN	fps cap	72	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	72	window60	89.1	89.1	99.2	99.2	0.775	0.775
E-DDQN	fps restore	90	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	rot jitter	0.02	whole	90.4	92.7	91.9	91.4	0.726	0.671
E-DDQN	rot jitter	0.02	window60	91.4	92.7	100.0	100.0	0.734	0.702
E-DDQN	rot jitter	0.05	whole	90.4	93.9	91.9	91.2	0.726	0.644
E-DDQN	rot jitter	0.05	window60	83.2	82.1	88.5	88.5	0.870	0.889
E-DDQN	rot jitter	0.1	whole	90.4	94.0	91.9	92.2	0.726	0.648
E-DDQN	rot jitter	0.1	window60	91.0	84.7	88.5	91.4	0.708	0.854
E-DDQN	rot smooth	0.25	whole	90.4	88.0	91.9	91.8	0.726	0.777
E-DDQN	rot smooth	0.5	whole	90.4	88.9	91.9	92.3	0.726	0.760
E-DDQN	speed scale down	−0.5	whole	90.4	86.7	91.9	90.9	0.726	0.802
E-DDQN	speed scale down	−0.25	whole	90.4	88.2	91.9	91.7	0.726	0.772
E-DDQN	speed scale up	0.1	whole	90.4	91.4	91.9	92.0	0.726	0.704
E-DDQN	speed scale up	0.1	window60	91.3	92.0	98.4	98.4	0.748	0.734
E-DDQN	speed scale up	0.25	whole	90.4	92.8	91.9	91.8	0.726	0.672
E-DDQN	speed scale up	0.25	window60	87.0	88.3	99.8	99.9	0.845	0.817
E-DDQN	speed scale up	0.5	whole	90.4	93.5	91.9	90.7	0.726	0.650
E-DDQN	speed scale up	0.5	window60	94.0	94.0	91.0	90.9	0.612	0.611
HGB	accel spikes	10	whole	97.0	97.1	85.4	87.8	0.543	0.555
HGB	accel spikes	10	window60	99.3	99.3	86.1	88.5	0.507	0.521
HGB	accel spikes	20	whole	97.0	97.9	85.4	94.5	0.543	0.575
HGB	accel spikes	20	window60	91.2	92.8	81.8	95.3	0.670	0.724
HGB	accel spikes	40	whole	97.0	98.1	85.4	95.7	0.543	0.577
HGB	accel spikes	40	window60	99.5	100.0	90.1	97.6	0.473	0.500
HGB	fps cap	45	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	45	window60	93.9	93.9	89.0	89.2	0.609	0.610
HGB	fps cap	60	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	60	window60	97.5	97.5	92.3	92.4	0.648	0.649
HGB	fps cap	72	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	72	window60	99.4	99.4	88.4	88.6	0.471	0.472
HGB	fps restore	90	whole	97.0	97.0	85.4	86.9	0.543	0.551
HGB	rot jitter	0.02	whole	97.0	97.0	85.4	85.0	0.543	0.541
HGB	rot jitter	0.02	window60	99.4	99.4	87.9	87.7	0.482	0.481
HGB	rot jitter	0.05	whole	97.0	96.6	85.4	82.9	0.543	0.538
HGB	rot jitter	0.05	window60	91.6	90.2	84.6	82.0	0.703	0.710
HGB	rot jitter	0.1	whole	97.0	96.3	85.4	82.7	0.543	0.543
HGB	rot jitter	0.1	window60	91.3	89.9	84.0	77.4	0.676	0.664
HGB	rot smooth	0.25	whole	97.0	97.0	85.4	88.3	0.543	0.558
HGB	rot smooth	0.5	whole	97.0	96.9	85.4	88.0	0.543	0.559
HGB	speed scale down	−0.5	whole	97.0	81.0	85.4	64.4	0.543	0.780
HGB	speed scale down	−0.25	whole	97.0	91.0	85.4	78.3	0.543	0.635
HGB	speed scale up	0.1	whole	97.0	97.3	85.4	88.9	0.543	0.556
HGB	speed scale up	0.1	window60	99.4	99.4	81.9	83.3	0.479	0.487
HGB	speed scale up	0.25	whole	97.0	98.0	85.4	90.7	0.543	0.550
HGB	speed scale up	0.25	window60	92.9	94.3	88.1	92.3	0.649	0.643
HGB	speed scale up	0.5	whole	97.0	98.2	85.4	91.7	0.543	0.551
HGB	speed scale up	0.5	window60	99.4	99.4	77.3	87.1	0.411	0.461
LogReg	accel spikes	10	whole	99.1	99.2	91.9	92.6	0.532	0.536
LogReg	accel spikes	10	window60	99.4	99.4	88.9	89.5	0.520	0.524
LogReg	accel spikes	20	whole	99.1	99.3	91.9	94.8	0.532	0.545
LogReg	accel spikes	20	window60	99.4	99.4	86.4	92.6	0.540	0.578
LogReg	accel spikes	40	whole	99.1	99.9	91.9	96.7	0.532	0.543
LogReg	accel spikes	40	window60	99.5	99.5	89.2	93.6	0.468	0.491
LogReg	fps cap	45	whole	99.1	97.0	91.9	86.3	0.532	0.549
LogReg	fps cap	45	window60	99.5	86.8	93.0	82.4	0.496	0.745
LogReg	fps cap	60	whole	99.1	98.1	91.9	88.4	0.532	0.535
LogReg	fps cap	60	window60	99.3	89.7	92.4	84.3	0.618	0.729
LogReg	fps cap	72	whole	99.1	98.6	91.9	90.2	0.532	0.534
LogReg	fps cap	72	window60	99.5	95.2	91.7	88.3	0.486	0.573
LogReg	fps restore	90	whole	99.1	99.1	91.9	91.9	0.532	0.532
LogReg	rot jitter	0.02	whole	99.1	99.2	91.9	92.1	0.532	0.533
LogReg	rot jitter	0.02	window60	99.5	99.5	91.5	91.6	0.499	0.500
LogReg	rot jitter	0.05	whole	99.1	99.3	91.9	92.5	0.532	0.533
LogReg	rot jitter	0.05	window60	96.4	96.8	92.2	92.5	0.673	0.668
LogReg	rot jitter	0.1	whole	99.1	99.2	91.9	92.7	0.532	0.535
LogReg	rot jitter	0.1	window60	95.7	96.9	91.1	91.3	0.629	0.605
LogReg	rot smooth	0.25	whole	99.1	99.1	91.9	91.7	0.532	0.532
LogReg	rot smooth	0.5	whole	99.1	99.1	91.9	91.7	0.532	0.532
LogReg	speed scale down	−0.5	whole	99.1	97.8	91.9	86.8	0.532	0.534
LogReg	speed scale down	−0.25	whole	99.1	98.7	91.9	89.2	0.532	0.526
LogReg	speed scale up	0.1	whole	99.1	99.2	91.9	92.9	0.532	0.536
LogReg	speed scale up	0.1	window60	99.4	99.4	87.8	88.7	0.513	0.518
LogReg	speed scale up	0.25	whole	99.1	99.4	91.9	93.6	0.532	0.536
LogReg	speed scale up	0.25	window60	99.4	99.4	90.8	91.9	0.520	0.527
LogReg	speed scale up	0.5	whole	99.1	99.7	91.9	94.9	0.532	0.537
LogReg	speed scale up	0.5	window60	99.5	99.5	86.0	92.5	0.452	0.486
RF	accel spikes	10	whole	98.7	99.0	81.7	84.5	0.485	0.494
RF	accel spikes	10	window60	99.3	99.3	75.0	79.1	0.444	0.467
RF	accel spikes	20	whole	98.7	99.7	81.7	96.2	0.485	0.543
RF	accel spikes	20	window60	97.6	99.1	68.6	96.2	0.466	0.605

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
RF	accel spikes	40	whole	98.7	99.8	81.7	97.4	0.485	0.549
RF	accel spikes	40	window60	99.5	100.0	81.1	99.8	0.427	0.511
RF	fps cap	45	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	45	window60	99.4	99.4	84.9	84.9	0.456	0.456
RF	fps cap	60	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	60	window60	99.2	99.2	84.7	84.7	0.569	0.569
RF	fps cap	72	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	72	window60	99.4	99.4	85.6	85.6	0.457	0.457
RF	fps restore	90	whole	98.7	98.8	81.7	81.8	0.485	0.483
RF	rot jitter	0.02	whole	98.7	98.2	81.7	79.4	0.485	0.482
RF	rot jitter	0.02	window60	99.4	99.4	85.0	83.0	0.467	0.456
RF	rot jitter	0.05	whole	98.7	97.7	81.7	79.0	0.485	0.491
RF	rot jitter	0.05	window60	96.1	91.5	84.3	82.4	0.625	0.691
RF	rot jitter	0.1	whole	98.7	98.0	81.7	81.1	0.485	0.496
RF	rot jitter	0.1	window60	96.3	92.8	82.1	79.1	0.563	0.616
RF	rot smooth	0.25	whole	98.7	99.0	81.7	85.9	0.485	0.502
RF	rot smooth	0.5	whole	98.7	99.1	81.7	84.9	0.485	0.494
RF	speed scale down	−0.5	whole	98.7	90.6	81.7	61.8	0.485	0.553
RF	speed scale down	−0.25	whole	98.7	95.9	81.7	73.3	0.485	0.501
RF	speed scale up	0.1	whole	98.7	99.1	81.7	84.3	0.485	0.492
RF	speed scale up	0.1	window60	99.0	99.0	75.5	77.5	0.452	0.464
RF	speed scale up	0.25	whole	98.7	99.1	81.7	86.7	0.485	0.504
RF	speed scale up	0.25	window60	98.6	98.3	84.0	85.5	0.502	0.515
RF	speed scale up	0.5	whole	98.7	99.0	81.7	92.1	0.485	0.537
RF	speed scale up	0.5	window60	99.4	99.4	66.3	85.6	0.354	0.453

S5 Post-hoc analyses

All analyses in this section were specified after the primary results were known. They use the splits, inner folds, selection objective, checkpoint and threshold rules, cross-fitting recipe and final seeds of the primary protocol, and no primary result was changed. They differ in when they were specified. The exposure-time baselines (S5.1), the Holm adjustment (S5.2), the comparison with always alarming (S5.3), the grouping sensitivity analyses (S5.4) and the cost curves (S5.5) were committed (`configs/posthoc.json`) before they were run. The paired comparisons of the exposure-time baselines with the primary models (main text, Section V-C) and the Q-value scale (S5.6) were added while the results were written up, without prior commitment. Sections S5.7–S5.10 were added after a pre-submission review, when all other results were known (key `review_round_2`); they use only the saved per-sample predictions, selected configurations and search logs. Section S5.11 was prompted by the same review but was specified, with its decision rules, and committed before any of its models was trained; like the exposure-time baselines of S5.1, it trains new models under the full protocol. As in the main text (Section III-G), bootstrap p-values between 0.01 and 0.05 in the tables of this section are approximate, and a difference whose confidence interval includes zero is a point estimate only.

S5.1 Exposure-time baselines: selected configurations

Time-LR searched all six values of C; Time+game HGB used the HGB search space and budget (16 configurations).

Model	Outer fold	Feature set	Selected hyper-parameters	Inner-OOF cost
Time-LR	0	time	C = 0.001	0.471
Time-LR	1	time	C = 1	0.528
Time-LR	2	time	C = 0.1	0.549
Time-LR	3	time	C = 0.01	0.497
Time-LR	4	time	C = 0.1	0.550
Time+game HGB	0	time+game	lr 0.02, iter 200, leaves 8, min leaf 300, ℓ_2 1	0.480
Time+game HGB	1	time+game	lr 0.02, iter 200, leaves 8, min leaf 300, ℓ_2 0	0.501
Time+game HGB	2	time+game	lr 0.02, iter 100, leaves 4, min leaf 300, ℓ_2 10	0.524
Time+game HGB	3	time+game	lr 0.02, iter 200, leaves 4, min leaf 300, ℓ_2 1	0.522
Time+game HGB	4	time+game	lr 0.02, iter 200, leaves 8, min leaf 300, ℓ_2 0	0.506

S5.2 Holm adjustment of the paired comparisons

Group-bootstrap comparisons of E-DDQN with the six other primary models under each operating rule, with Holm's step-down adjustment over the six comparisons of each rule.

Rule	B	ΔCost (E-DDQN−B)	95 % CI	p_{group}	p_{Holm}
D1 (min cost)	E-CB	+0.175	[+0.016, +0.448]	0.014	0.070
D1 (min cost)	DQN	+0.190	[−0.017, +0.493]	0.079	0.105
D1 (min cost)	CS-MLP	+0.190	[+0.005, +0.485]	0.039	0.105
D1 (min cost)	LogReg	+0.194	[+0.011, +0.477]	0.026	0.104
D1 (min cost)	HGB	+0.183	[+0.011, +0.460]	0.035	0.105
D1 (min cost)	RF	+0.242	[+0.052, +0.528]	0.001	0.006
D0 ($\hat{p} \geq 1/6$)	E-CB	+0.214	[+0.025, +0.487]	0.018	0.054
D0 ($\hat{p} \geq 1/6$)	DQN	+0.242	[+0.039, +0.542]	0.012	0.048
D0 ($\hat{p} \geq 1/6$)	CS-MLP	+0.204	[−0.030, +0.507]	0.113	0.226
D0 ($\hat{p} \geq 1/6$)	LogReg	+0.087	[−0.293, +0.541]	0.694	0.694
D0 ($\hat{p} \geq 1/6$)	HGB	+0.328	[+0.078, +0.639]	0.003	0.015
D0 ($\hat{p} \geq 1/6$)	RF	+0.411	[+0.147, +0.748]	0.001	0.006
val. recall ≥0.95	E-CB	+0.220	[+0.017, +0.566]	0.021	0.080
val. recall ≥0.95	DQN	+0.236	[+0.013, +0.580]	0.028	0.080
val. recall ≥0.95	CS-MLP	+0.294	[+0.033, +0.652]	0.013	0.065
val. recall ≥0.95	LogReg	+0.307	[+0.027, +0.687]	0.020	0.080
val. recall ≥0.95	HGB	+0.314	[+0.030, +0.711]	0.023	0.080
val. recall ≥0.95	RF	+0.342	[+0.067, +0.719]	0.003	0.018

S5.3 Comparison with always alarming

Paired bootstrap of each model's D1 cost (seed ensemble) minus the cost of always alarming; Holm adjustment over the nine models (group units).

Model	ΔCost (model−always)	CI (groups)	p	p_{Holm}	CI (sessions)	p
E-DDQN	+0.168	[−0.024, +0.466]	0.104	0.728	[−0.026, +0.438]	0.116
E-CB	−0.007	[−0.093, +0.103]	0.810	1.000	[−0.089, +0.105]	0.800
DQN	−0.022	[−0.104, +0.062]	0.560	1.000	[−0.099, +0.075]	0.598
CS-MLP	−0.022	[−0.046, −0.002]	0.037	0.296	[−0.049, +0.004]	0.109
LogReg	−0.026	[−0.061, +0.006]	0.136	0.816	[−0.058, +0.005]	0.113
HGB	−0.015	[−0.122, +0.121]	0.765	1.000	[−0.104, +0.092]	0.744
RF	−0.074	[−0.135, −0.019]	0.012	0.108	[−0.120, −0.025]	0.002
Time-LR	−0.010	[−0.073, +0.080]	0.737	1.000	[−0.080, +0.080]	0.782
Time+game HGB	−0.041	[−0.100, +0.051]	0.302	1.000	[−0.099, +0.048]	0.275

S5.4 Grouping sensitivity

Paired bootstrap resampling sessions instead of linkage groups (Holm over the six primary comparisons), and the range of the cost difference in a leave-one-group-out jackknife over the 21 groups.

B	ΔCost (E-DDQN−B)	CI (sessions)	p	p_{Holm}	jackknife range	sign flips
E-CB	+0.175	[+0.014, +0.423]	0.014	0.070	+0.068 to +0.201	0
DQN	+0.190	[+0.001, +0.454]	0.050	0.087	+0.084 to +0.248	0
CS-MLP	+0.190	[+0.006, +0.452]	0.037	0.087	+0.084 to +0.225	0
LogReg	+0.194	[+0.007, +0.454]	0.029	0.087	+0.094 to +0.225	0
HGB	+0.183	[+0.014, +0.440]	0.021	0.084	+0.076 to +0.215	0
RF	+0.242	[+0.055, +0.503]	0.001	0.006	+0.137 to +0.284	0
Time-LR	+0.178	[−0.015, +0.439]	0.089	–	+0.075 to +0.213	0
Time+game HGB	+0.209	[+0.016, +0.470]	0.030	–	+0.101 to +0.248	0

S5.5 Validation-chosen cost curves

Test cost of the seed ensemble divided by the cost of the better trivial policy, for each false-negative to false-positive ratio R (threshold chosen on inner-validation predictions at each R). At $R = 5$ the values reproduce the D1 seed-ensemble costs exactly (checked programmatically).

Model	1	1.5	2	3	4	5	7	10	15	20
E-DDQN	1.004	0.921	1.034	1.130	1.241	1.301	1.382	1.322	1.448	1.508
E-CB	0.828	0.819	0.900	0.920	0.954	0.988	1.068	0.992	1.007	1.013
DQN	0.921	0.796	0.865	0.845	0.929	0.961	1.013	1.068	1.208	1.304
CS-MLP	0.846	0.773	0.835	0.980	0.960	0.961	0.969	0.981	0.985	0.985
LogReg	0.852	0.868	0.915	0.941	0.946	0.953	0.966	0.983	0.989	1.010
HGB	0.897	0.770	0.808	0.884	0.945	0.973	1.001	0.980	0.991	0.998
RF	0.861	0.717	0.773	0.836	0.864	0.868	0.893	0.928	0.960	0.983
Time-LR	0.843	0.718	0.781	0.840	0.883	0.982	0.990	0.984	1.000	1.000
Time+game HGB	0.816	0.731	0.760	0.833	0.882	0.926	0.942	1.014	1.053	1.064

S5.6 Q-value scale

Largest absolute Q-value over the final runs of each outer fold relative to the return bound $r_{\max}/(1-\gamma)$ of the selected discount ($r_{\max} = 0.5$ for the asymmetric reward, 1 for the symmetric reward).

Model	Outer fold	γ	max $\lvert Q\rvert$	bound $r_{\max}/(1-\gamma)$	ratio
E-DDQN	0	0.99	41.45	50.00	0.83
E-DDQN	1	0.99	61.78	50.00	1.24
E-DDQN	2	0.99	35.08	50.00	0.70
E-DDQN	3	0.99	4.13	50.00	0.08
E-DDQN	4	0.9	11.56	5.00	2.31
E-CB	0	0	0.70	0.50	1.41
E-CB	1	0	0.68	0.50	1.37
E-CB	2	0	0.68	0.50	1.36
E-CB	3	0	0.89	0.50	1.79
E-CB	4	0	0.95	0.50	1.89
DQN	0	0	2.24	1.00	2.24
DQN	1	0.9	4.09	10.00	0.41
DQN	2	0.9	3.05	10.00	0.30
DQN	3	0.9	10.10	10.00	1.01
DQN	4	0	2.08	1.00	2.08

S5.7 Within-fold ROC-AUC

Pooled ROC-AUC combines the test scores of five fold-specific models and therefore mixes discrimination with differences in score scale between folds. Within-fold ROC-AUC is computed in each outer-test fold and averaged over folds (seed mean: also averaged over the five seeds).

Model	Pooled (seed mean)	Within-fold (seed mean)	Pooled (ensemble)	Within-fold (ensemble)
E-DDQN	0.538	0.539	0.541	0.552
E-CB	0.653	0.658	0.659	0.665
DQN	0.661	0.666	0.665	0.666
CS-MLP	0.644	0.666	0.654	0.672
LogReg	0.626	0.636	0.626	0.636
HGB	0.692	0.702	0.692	0.702
RF	0.688	0.715	0.688	0.715
Time-LR	0.704	0.709	0.704	0.709
Time+game HGB	0.676	0.699	0.676	0.699

S5.8 Exact group sign-flip tests

Two-sided sign-flip permutation test of the pooled D1 cost difference of the seed ensembles, with the 21 linkage groups as units and all 2^{21} sign patterns enumerated. Holm adjustment over the six primary comparisons with E-DDQN and over the nine comparisons with always alarming. The group-bootstrap p-values of Section S5.2 (rule D1) are shown for comparison.

Comparison (D1, seed ensembles)	ΔCost	p_{boot}	$p_{\text{boot,Holm}}$	$p_{\text{sign-flip}}$	$p_{\text{sign-flip,Holm}}$
E-DDQN vs E-CB	+0.175	0.014	0.070	0.048	0.241
E-DDQN vs DQN	+0.190	0.079	0.105	0.127	0.330
E-DDQN vs CS-MLP	+0.190	0.039	0.105	0.116	0.330
E-DDQN vs LogReg	+0.194	0.026	0.104	0.094	0.330
E-DDQN vs HGB	+0.183	0.035	0.105	0.083	0.330
E-DDQN vs RF	+0.242	0.001	0.006	0.014	0.083
E-DDQN vs Time-LR	+0.178	–	–	0.117	–
E-DDQN vs Time+game HGB	+0.209	–	–	0.067	–
Time-LR vs RF	+0.063	0.021		0.044	
Time+game HGB vs RF	+0.032	0.274		0.302	
E-DDQN vs always alarm	+0.168			0.250	1.000
E-CB vs always alarm	−0.007			0.922	1.000
DQN vs always alarm	−0.022			0.620	1.000
CS-MLP vs always alarm	−0.022			0.053	0.422
LogReg vs always alarm	−0.026			0.172	1.000
HGB vs always alarm	−0.015			0.769	1.000
RF vs always alarm	−0.074			0.022	0.194
Time-LR vs always alarm	−0.010			0.820	1.000
Time+game HGB vs always alarm	−0.041			0.358	1.000

S5.9 Selected feature sets and within-fold ROC-AUC

Exploratory. Feature set selected on validation data for each primary model and outer fold (T: telemetry; T+Q: telemetry and questionnaire indicators; E: exposure; E+Q: exposure and questionnaire indicators) and the within-fold ROC-AUC of that model on the fold's test groups (mean over seeds). Feature sets were selected, not assigned, so the association does not establish a cause.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
E-DDQN	E+Q 0.541	E+Q 0.505	E+Q 0.723	E+Q 0.450	E+Q 0.476
E-CB	E 0.642	E 0.717	E 0.748	E 0.737	E+Q 0.446
DQN	E 0.706	E+Q 0.530	E 0.843	E 0.736	E+Q 0.515
CS-MLP	E+Q 0.562	E+Q 0.607	E 0.812	T 0.702	T 0.644
LogReg	E 0.623	T+Q 0.428	E 0.810	E 0.712	E+Q 0.610
HGB	E 0.631	E 0.740	E 0.791	T 0.761	E 0.585
RF	T 0.630	E 0.728	E 0.771	E 0.761	T+Q 0.683

S5.10 Search draw

Feature sets among each family's random-search candidates (T: telemetry; T+Q: telemetry and questionnaire indicators; E: exposure; E+Q: exposure and questionnaire indicators). The candidates were drawn once per family with a fixed seed and reused in every outer fold.

Model	Candidates	T	T+Q	E	E+Q
E-DDQN	16	4	2	2	8
E-CB	16	3	3	6	4
DQN	16	2	5	6	3
CS-MLP	16	4	5	5	2
LogReg	12	3	3	4	2
HGB	16	4	4	3	5
RF	16	4	3	4	5

For E-DDQN, the lowest pooled inner-validation 5:1 cost among candidates with (Q) and without the questionnaire indicators in each outer fold; the selection rule picks the lower value.

E-DDQN, outer fold	0	1	2	3	4
lowest validation cost, candidates with Q	0.424	0.452	0.484	0.490	0.483
lowest validation cost, candidates without Q	0.487	0.483	0.521	0.518	0.542

S5.11 Search-draw sensitivity (specified before it was run)

Prompted by the dependence of E-DDQN's result on its single random draw (S5.10), this analysis was specified in `configs/posthoc.json` (key `search_draw_sensitivity`), together with its decision rules, and committed before any of its models was trained. E-DDQN and the standard DQN were each tuned again under the full protocol—same splits, inner folds, search training seed, budget of 16 configurations, selection rule, checkpoint rule, cross-fitting, five final seeds and threshold rules—with two further random draws from the full search space (configuration seeds 9120 and 9130 for E-DDQN, 9122 and 9132 for DQN) and with one draw of 16 configurations restricted to the exposure feature set (9200 and 9202). Perturbation variants were not scored. Draw 1 is the primary analysis and is reproduced exactly. Every variant is reported; none replaces the primary results. The decision rules were: (R1) the statement that E-DDQN does not outperform the standard DQN is challenged if, in any pair, E-DDQN's seed-ensemble cost is lower and the 95 % group-bootstrap interval of the difference excludes zero; (R2) E-DDQN's result is called draw-dependent if its seed-mean cost or its within-fold ROC-AUC spans more than 0.05 over its three full-space draws; (R3) under the exposure-only restriction, a within-fold ROC-AUC of E-DDQN within 0.03 of DQN's supports the feature-set explanation of Section V-G (iii), one more than 0.05 below DQN's means that this explanation is insufficient, and anything in between is inconclusive.

Per variant: number of exposure-only candidates in the draw, number of outer folds whose selected configuration uses the questionnaire indicators (Q folds), 5:1 cost (rule D1; mean ± SD over seeds), seed-ensemble cost, pooled and within-fold ROC-AUC (seed means), and the discount factor selected in each outer fold.

Variant	E-only cand.	Q folds	$\text{Cost}_{5:1}$ (seed mean)	Ens. cost	ROC-AUC	Within-fold AUC	Selected γ
E-DDQN, draw 1 (primary)	2	5/5	0.663±0.048	0.726	0.538	0.539	0.99, 0.99, 0.99, 0.99, 0.9
DQN, draw 1 (primary)	6	2/5	0.584±0.040	0.536	0.661	0.666	0.0, 0.9, 0.9, 0.9, 0.0
E-DDQN, draw 2	8	0/5	0.549±0.014	0.527	0.711	0.714	0.99, 0.99, 0.9, 0.99, 0.0
DQN, draw 2	4	3/5	0.634±0.066	0.662	0.634	0.631	0.9, 0.0, 0.0, 0.5, 0.0
E-DDQN, draw 3	7	2/5	0.558±0.019	0.561	0.658	0.671	0.9, 0.9, 0.0, 0.0, 0.99
DQN, draw 3	3	4/5	0.580±0.034	0.558	0.628	0.629	0.9, 0.9, 0.9, 0.5, 0.9
E-DDQN, exposure only	16	0/5	0.565±0.021	0.520	0.676	0.687	0.9, 0.5, 0.9, 0.9, 0.5
DQN, exposure only	16	0/5	0.536±0.017	0.545	0.705	0.712	0.5, 0.0, 0.9, 0.5, 0.0

Paired comparisons of E-DDQN with the standard DQN within each draw (seed ensembles, rule D1; 2,000 group-bootstrap replicates; exact sign-flip test over all 2^{21} sign patterns; ROC-AUC difference of the seed-ensemble scores).

Draw	ΔCost (E-DDQN−DQN)	95 % CI	p_{boot}	$p_{\text{sign-flip}}$	ΔROC-AUC	95 % CI	p_{boot}
draw 1 (primary)	+0.190	[−0.017, +0.493]	0.079	0.127	−0.125	[−0.264, −0.018]	0.021
draw 2	−0.135	[−0.395, +0.024]	0.167	0.279	+0.088	[+0.021, +0.162]	0.009
draw 3	+0.004	[−0.052, +0.061]	0.953	0.907	+0.031	[−0.029, +0.115]	0.354
exposure only	−0.025	[−0.061, +0.025]	0.276	0.329	−0.025	[−0.066, +0.009]	0.156

Outcome under the pre-specified rules. R1 (headline challenged): no; no pair gave E-DDQN a significantly lower cost (differences −0.135 to +0.190; smallest group-bootstrap $p = 0.079$). R2 (draw-dependent): yes; over its three full-space draws E-DDQN's seed-mean cost spans 0.114 (0.549–0.663) and its within-fold ROC-AUC 0.175 (0.539–0.714), against 0.580–0.634 and 0.629–0.666 for the standard DQN. R3 (feature-set explanation): supported; with the exposure features only, the standard DQN's within-fold ROC-AUC exceeds E-DDQN's by 0.026. Beyond the rules, the ROC-AUC differences show that the ranking of the two agents reverses between draws: the standard DQN ranks seconds better in draw 1 ($p = 0.021$, approximate) and E-DDQN in draw 2 ($p = 0.009$). The variants were not compared with the other models by any test. The clean rerun of the final version reproduced every variant's per-sample predictions exactly (Section S7).

S6 Reconstruction of the earlier protocol

An earlier, unpublished version of this study reported 96.88 % accuracy and 99.53 % recall for E-DDQN. Its 41-feature input files, preprocessing code and trained models were not available, so the following is a reconstruction of its protocol, not an exact reproduction. The earlier version's partition sizes (14,680 / 3,146 / 3,146; 20,972 in total; 3,670 training samples per grade) are reproduced exactly by oversampling every discomfort grade of the curated release to the size of grade 0 (5,243) and then splitting at row level.

Model	Evaluation protocol	Acc.	Recall	Spec.	MCC	ROC-AUC
HGB	Oversample to 5,243/grade, then 70/15/15 row split (earlier protocol)	97.6	99.8	91.0	0.935	0.995
HGB	Row-level random split, no oversampling	95.3	94.1	96.3	0.905	0.992
HGB	Grouped 5-fold split (session-linkage groups)	52.2	33.6	66.9	0.005	0.523
E-DDQN (earlier code)	Earlier script, earlier protocol (proxy)	95.2	99.2	83.1	0.870	–
E-DDQN (earlier code)	Earlier script, grouped split (proxy)	41.1	92.1	7.7	−0.004	–

Under this protocol 2,170 of the 2,359 positive test rows (92 %) have an exact duplicate in the training split, and all sessions occur in both splits.

Implementation pitfalls found in the earlier code

The following defects of the earlier E-DDQN implementation are guarded against by the unit tests of the present code (Section S7); they are listed because they are easy to make whenever a classifier is trained as a Q-learner.

1. The next state was the next row of a shuffled chunk, with no terminal flags, so $\gamma = 0.99$ bootstrapped across unrelated samples.
2. The true label was stored as the action together with the reward of the random action actually taken (a form of teacher forcing).
3. The Polyak update ignored batch-normalization buffers, so the target network normalized with mean 0 and variance 1 throughout training.
4. The Double-DQN arg-max ran the online network in training mode.
5. The test set was scored during training (every 10 episodes), which invites test-set monitoring.
6. SHAP attributed $Q(s, 1)$, which contains $V(s)$, on 50 test rows with the additivity check disabled.

S7 Reproducibility

- Software: Python 3.13.16, PyTorch 2.14.1 (CPU), scikit-learn 1.9.1, NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, SHAP 0.53.0, statsmodels 0.15.0 (`requirements.txt`).
- One command regenerates everything from the raw XML files: `bash run_all.sh --fresh` (about 2.5 to 3.5 hours of wall time on four CPU cores, dominated by the reinforcement-learning hyper-parameter searches and ablation). A clean rerun of the final version (commit `2e49e93`, which differs from this version only in these reproducibility statements) from an empty directory, including the search-draw analysis, took about 2 hours 46 minutes of computing and reproduced every per-sample prediction, every reported number and the text of the paper and supplement exactly; it was interrupted once by a restart of the computing container and resumed from its saved per-job outputs, as recorded in `results/qc/clean_rerun.md`.
- The post-hoc analyses (Section S5) are recorded in `configs/posthoc.json`; Section S5 states which of them were committed before they were run and which were added later. They are produced by `scripts/11_posthoc.py` (`run` and `analyze`; seeds 801–802 for the post-hoc search, bootstrap seeds 5–7), except the search-draw sensitivity analysis (Section S5.11, key `search_draw_sensitivity`), which is produced by `scripts/13_search_draw_sensitivity.py` (configuration seeds 9120, 9130, 9122, 9132, 9200 and 9202; bootstrap seeds 5 and 6).
- Seeds: split 20261012 (outer) and 20261013–20261017 (inner); hyper-parameter search 777 (+family index for configuration sampling); final training 1000–1004; perturbations 4242; bootstrap 11 (CIs) and 5 (paired tests).
- Every number in the manuscript and in this supplement is produced by `scripts/05_make_tables.py` (primary analysis) from `results/tables/*.csv`, which `scripts/04_aggregate.py` computes from the per-sample predictions in `results/predictions/`, or by `scripts/11_posthoc.py` and `scripts/13_search_draw_sensitivity.py` (post-hoc analyses, Section S5). `scripts/08_consistency_check.py` recomputes confusion matrices, metrics and McNemar statistics with independent library implementations and checks them against the manuscript macros.
- Executable sanity tests of the corrected agent: `scripts/00_sanity_tests.py` (reward regret ratio, implied threshold, within-session transitions, terminal flags, reward/action consistency of replay, target-network buffer synchronisation, Q-value bounds, McNemar identity).